TQM 2020
Econometric
Time series
Multivariate

77

Multivariate

sequential variance Maximization
% of PC-I

Preservation of T-var

$\Sigma = \Sigma\sigma_{11} + \sigma_{22} = 1 + 100 = 101$

$Y_1 \& Y_2 = \lambda_1 + \lambda_2 = 100.16 + 0.84 = 101$

Tot var(X) = T.V(Y)

Orthogonality of Loading vector

$$e_1' e_2 = \begin{bmatrix} 0.0399 & 0.9019 \end{bmatrix} \begin{bmatrix} 0.0399 \\ 0.9019 \end{bmatrix}$$

$$e_1' e_2 = 0.0399 - 0.9019$$

Uncorrelationess of PC's

$$cov(Y_1, Y_2) = cov(e_1' x \quad e_2' x)$$
$$= e_1' \, cov(X) \, e_2$$
$$= e_1' \Sigma e_2$$

$$\begin{bmatrix} 0.0399 & 0.9992 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 4 & 100 \end{bmatrix} \begin{bmatrix} 0.9992 \\ 0.0399 \end{bmatrix}$$

$$" \quad \begin{bmatrix} 0.8396 \\ 0.0068 \end{bmatrix} =$$

$$0.02 \cong 0$$

-: Past Paper 2014:-

Question No. 2:-

Differentiate Between

(i)

Principle Component Analysis and Factor Analysis:-

Principle Component Analysis:-

Principle components are the linear combinations of the "p" random variables $X_1, X_2, \cdots, X_p$.

Geometrically, these linear combination represents the selection of a new co-ordinate system obtained by rotating system with $X_1, X_2, \cdots, X_p$ as the co-ordinate axis.

Factor Analysis:-

The essential purpose of factor analysis is to describe if possible the covariance relationship among many variables in terms of a few underlying but unobserveable random quantities called factor OR Latent variable or unobserved or Abstract variable.

(ii)

Communalities and Factor Loadings:-

Communalities:-

The portion of the variance of the i^{th} variable contributed by the 'm' common factor is called i^{th} communals.

Factor Loading:-

Factor loading is basically the correlation coefficient for the variable and factor. Factor loading shows the variance explained by the variable on that particular factor.

(3)

Common Factor and Specific Factor:-

Common Factor:-

The essential purpose of factor analysis is to describe if possible the covariance relationship among many variables in terms of a few underlying but unobserveable random quantities called factors OR Latent variable or Unobserved or abstract variable.

Specific Factor:-

A specific factor is one which is stuck in an industry or is immobile between industries in response to changes in market conditions.

The specific factor model assumes that an economy produces two goods using two factors of production, capital and labor, in a perfectly competitive market.

(4)

Discriminant Analysis and Canonical Correlations:-

Discriminant Analysis:-

To describe either graphically or algebrically the differential features of objects from several known correlations (populations). we tried to find discriminants whose numerical values are such that the collections are seperated, as much as possible.

Canonical Correlations:-

As multiple correlation "R" is a measure of linear relationship between 1 "y" and several "x's".

Canonical correlation is a measure of linear relationship between several "y's" and several "x's".

Mahalanobis Distance and Cook's distances:-

Mahalanobis Distance:-

The Mahalanobis distance is the distance between two points in multivariate space. In a regular Euclidean space, variables (e.g, x, y, z) are represented by axes drawn at right angles to each other; The distance between any two points can be measured with a ruler.

Cook's Distance:-

In statistics, Cook's distance or Cook's D is a commonly used estimate of the influence of a data point when performing a least-squares regression analysis.

It is named after the American statistician R Dennis Cook, who introduced the concept in 1977.

-: Past Paper 2015:-

Question No. 2:-

(i)

Factor Rotations:-

Significant loadings should be much greater than 0.30.

Insignificant loadings should be must less than 0.30.

Idealy, we should like to see a pattern of loadings such that each variable loads highly ($l_{ij} > 0.3$) on 1 factor and has small to moderate loadings on other factors.

This rotation could be done in two ways.

Orthogonal Vario Rotation (Varimax)
Oblique Rotation.

(ii)

Discriminant Analysis:-

Repeat

(iii)

Canonical Correlations Analysis:-

Repeat

(iv)

Principal Component Analysis:-

Repeat

:- Past Paper 2016 :-

(i)

~~Prince~~

All shorts repeat see 2014.

:- Past Paper 2017 :-

All shorts repeat see 2014.

:- Past Paper 2019 :-

Question No.2 :-

(i)

Common Factor and Specific Fator :-

Repeat

(ii)

Fisher's Discrimination Rule :-

Fisher's approach to the discrimination problem is based on the data matrix "$\underset{\sim}{X}$" with no assumption on the parametric form of population $P_1, P_2, \ldots\ldots, P_g$ it looks for the linear function.

$$y = \underline{a}' \underline{x}$$

which minimize the ratio of b/w group sum of sq's to the within group sums of square.

(3)

Canonical Correlation Analysis:-

Repeat

(4)

Principal Component Analysis for Standardized variables:-

Repeat

:Past Paper 2014:-

Question No. 3:-

Derive the distribution of Hotelling T^2- statistic for testing $H_0: \mu = \mu_0$.

Solution:-

Hotelling's T^2- Test:-

$$\left.\begin{aligned} H_0 &: \underline{\mu}_{p\times 1} = \underline{\mu}_0 \\ H_1 &: \underline{\mu}_{p\times 1} \neq \underline{\mu}_0 \end{aligned}\right\} \longrightarrow (A)$$

let $\underline{y}_1, \underline{y}_2, \ldots, \underline{y}_n \sim N_p(\underline{\mu}, \underset{\sim}{\Sigma})$

where "$\underset{\sim}{\Sigma}$" is unknown.

Univariate Case:-

$$H_0: \mu = \mu_0$$
$$H_1: \mu \neq \mu_0$$

$Y_1, Y_2, \ldots, Y_n \sim N(\mu, \sigma^2)$ where σ^2 is unknown

$$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$$

Squaring on both sides.

$$t^2 = \frac{(\bar{y}-\mu_0)^2}{s^2/n}$$

$$t^2 = \frac{n(\bar{y}-\mu_0)^2}{s^2}$$

$$t^2 = n(\bar{y}-\mu_0)'(s^2)^{-1}(\bar{y}-\mu_0)$$ ∵ As $(\bar{y}-\mu_0)$ is a smaler quantity

$$\Rightarrow t^2 = n(\bar{y}-\mu_0)'(s^2)^{-1}(\bar{y}-\mu_0)$$

Now considering its MVT case:-

$$T^2 = n(\underline{\bar{y}}-\underline{\mu_0})' \underset{\sim}{S}^{-1}(\underline{\bar{y}}-\underline{\mu_0}) \sim T^2_{(\alpha)(p,n-1)}$$

Ho in eq (A) will be rejected if

$$T^2 \geq T^2_{\alpha(p,n-1)}$$ ∵ p = No. of variables & n = No. of observation

The key assumption in the T^2 dist is the independence of $\underline{\bar{y}}$ and $\underset{\sim}{S}$.

This assumption automatically holds when sampling is from MVT Normal population.

Distribution of T^2:-

In univariate case:-

$$Z \sim N(0,1)$$
$$W \sim \chi^2(v)$$

Then by definition

$$t = \frac{z}{\sqrt{W/V}}$$

taking square on both sides.

$$t^2 = \frac{z^2}{W/V}$$

$$t^2 = z'\left(\frac{W}{V}\right)^{-1} z$$

Now In multivariate case:

$$\underline{z} \sim N_p(0, \underset{\sim}{\Sigma})$$
$$\underset{\sim}{W} \sim W_p(V, \underset{\sim}{\Sigma})$$

Also $\underline{z}$ and $\underset{\sim}{W}$ are independent. Then

$$T^2 = \underline{z}'\left(\frac{\underset{\sim}{W}}{V}\right)^{-1}\underline{z}$$

$$T^2 = \frac{\underline{z}'\underset{\sim}{W}^{-1}\underline{z}}{V^{-1}} = V[\underline{z}'\underset{\sim}{W}^{-1}\underline{z}]$$

$$\frac{T^2}{V} = \underline{z}'\underset{\sim}{W}^{-1}\underline{z} \Rightarrow \frac{T^2}{V} = \frac{1}{1/\underline{z}'\underset{\sim}{W}^{-1}\underline{z}}$$

Multiply and divide by $\underline{z}'\underset{\sim}{\Sigma}^{-1}\underline{z}$

$$\frac{T^2}{V} = \frac{\underline{z}'\underset{\sim}{\Sigma}^{-1}\underline{z}}{\underline{z}'\underset{\sim}{\Sigma}^{-1}\underline{z}/\underline{z}'\underset{\sim}{W}^{-1}\underline{z}}$$

$$\therefore \quad Q.F = \underline{z}'\, \underset{\sim}{\Sigma}^{-1}\, \underline{z} \sim \chi^2(p)$$

Also

$$\therefore \quad \frac{\underline{z}'\,\Sigma^{-1}\,\underline{z}}{\underline{z}'\,\underset{\sim}{W}^{-1}\,\underline{z}} \sim \chi^2(v-p+1)$$

Now;

$$\frac{T^2}{v}\left(\frac{v-p+1}{p}\right) = \frac{\underline{z}'\,\Sigma^{-1}z/p}{\left(z'\Sigma^{-1}z/z'\underset{\sim}{W}^{-1}z\right)/(v-p+1)}$$

$$\frac{T^2}{v}\left(\frac{v-p+1}{p}\right) = \frac{\chi^2(p)}{\chi^2(v-p+1)}$$

So;

$$\frac{T^2}{v}\left(\frac{v-p+1}{p}\right) \sim F(\alpha)_{(p,\,v-p+1)}$$

$$\Rightarrow \quad T^2 = \left(\frac{vp}{v-p+1}\right) F(\alpha)_{(p,\,v-p+1)}$$

Question No. 4:-

Consider the sample correlation matrix based on 150 observations:

$$R = \begin{array}{cc} & \text{Row} \\ \text{column} & \begin{bmatrix} 1.00 & 0.75 & 0.65 & 0.66 \\ 0.75 & 1.00 & 0.69 & 0.73 \\ 0.65 & 0.69 & 1.00 & 0.67 \\ 0.66 & 0.73 & 0.67 & 1.00 \end{bmatrix} \end{array}$$

Test the hypothesis that

$$H_0: \rho = \begin{bmatrix} 1 & \rho & \rho & \rho \\ & 1 & \rho & \rho \\ & & 1 & \rho \\ & & & 1 \end{bmatrix}$$

Solution:-

(i) Hypotheses:-

$$H_0: \underset{\sim}{\rho} = \underset{\sim}{\rho}_0$$
$$H_1: \underset{\sim}{\rho} \neq \underset{\sim}{\rho}_0$$

(ii) Level of significance:-

$$\alpha = 0.05$$

(iii) Test Statistic:-

sample correlation Matrix

$$T = \frac{n-1}{(1-\bar{r})^2}\left[\sum_{i<k}\sum (r_{ik}-\bar{r})^2 - \hat{\gamma}\sum_{k=1}^{p}(\bar{r}_k-\bar{r})^2\right]$$

(iv) Computations:-

$$\bar{r}_k = \frac{1}{p-1}\sum_{\substack{i=1 \\ i\neq k}}^{p}(r_{ik})$$

$$\bar{r}_1 = \frac{1}{p-1}\sum_{\substack{i=1 \\ i\neq k}}^{4} r_{i1}$$

$$\bar{r}_1 = \frac{1}{4-1}(0.75+0.65+0.66) =$$

$$\bar{r}_1 = 0.6867$$

$$\bar{r}_2 = \frac{1}{4-1}(0.75 + 0.69 + 0.73).$$

$$\bar{r}_2 = 0.7233$$

$$\bar{r}_3 = \frac{1}{4-1}(0.65 + 0.69 + 0.67)$$

$$\bar{r}_3 = 0.6633 = 0.67$$

$$\bar{r}_4 = \frac{1}{4-1}(0.66 + 0.73 + 0.67)$$

$$\bar{r}_4 = 0.6867$$

$$\bar{r} = \frac{\bar{r}_1 + \bar{r}_2 + \bar{r}_3 + \bar{r}_4}{4}$$

$$\bar{r} = \frac{0.6867 + 0.7233 + 0.6633 + 0.6867}{4}$$

3.8078

$$\boxed{\bar{r} = 0.69}$$

8.1351

~~0.7688~~

0.1922

$$\hat{\gamma} = \frac{(p-1)^2\left(1-(1-\bar{r})^2\right)}{p-(p-2)(1-\bar{r})^2}$$

0.31

0.0961

0.9039

$$\hat{\gamma} = \frac{(4-1)^2\left(1-(1-0.69)^2\right)}{4-(4-2)(1-0.69)^2}$$

$$\boxed{\hat{\gamma} = 2.1364}$$

$$\sum_{i<k}\sum (r_{ik} - \bar{r})^2 = (0.75-0.69)^2 + (0.65-0.69)^2 + (0.66-0.69)^2 + (0.69-0.69)^2 + (0.73-0.69)^2 + (0.67-0.69)^2$$

$$= 0.0081$$

$$\sum_{k=1}^{p} (\bar{r}_k - \bar{r})^2 = (0.6867 - 0.69)^2 + (0.7233 - 0.69)^2 + (0.6733 - 0.69)^2 + (0.6867 - 0.69)^2$$

$$\sum_{k=1}^{p} (\bar{r}_k - \bar{r})^2 = 0.0018$$

$$T = \frac{(150-1)}{(1-0.69)^2} \left[(0.0081) - (2.1364)(0.0018) \right]$$

$$\boxed{T = 7.5964}$$

(5) Critical Region:-

Reject H_0 if

$$T > \chi^2_{(\alpha)} \frac{(p-2)(p+1)}{2}$$

$$T > \chi^2_{0.05}(5)$$

$$T > 11.07$$

(6) Decision:-

As $7.5964 < 11.07$, so do no reject H_0 at 5% level of significance. i.e $\underset{\sim}{\rho} \neq \underset{\sim}{\rho}_0$

Question No. 5:-

The following mean vector

and covariance matrices are based on $n_1 = n_2 = 100$ observations:

$$\bar{X}_1 = \begin{bmatrix} 6.213 \\ 3.133 \end{bmatrix}, \quad \bar{X}_2 = \begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}, \quad \underset{\sim}{S_1} = \begin{bmatrix} 1.813 & 0.321 \\ 0.321 & 0.937 \end{bmatrix}$$

$$\underset{\sim}{S_2} = \begin{bmatrix} 2.193 & 1.654 \\ & 3.789 \end{bmatrix}.$$

1.654

Find the Fisher's Linear Discriminant function and Discriminant rule. Also allocate the new observation $X = [7.2 \quad 3.1]'$ to any these populations.

Solution:-

The Fisher's Linear Discriminant Function is:

$$y = \underline{a}'\underline{x}$$

Since, it is a two group case: so;

$$\underline{a} = \underset{\sim}{W}^{-1}(\bar{x}_1 - \bar{x}_2)$$

$\underset{\sim}{u}$ <

$$\underset{\sim}{W} = \sum_{g=1}^{n} n_g S_g$$

$\underset{\sim}{W}^{-1}$ Adj

$$\underset{\sim}{W} = n_1 \underset{\sim}{S_1} + n_2 \underset{\sim}{S_2}$$

$$\underset{\sim}{W} = 100 \begin{bmatrix} 1.813 & 0.321 \\ 0.321 & 0.937 \end{bmatrix} + 100 \begin{bmatrix} 2.193 & 1.654 \\ 1.654 & 3.789 \end{bmatrix}$$

$$\underset{\sim}{W} = \begin{bmatrix} 181.3 & 32.1 \\ 32.1 & 93.7 \end{bmatrix} + \begin{bmatrix} 219.3 & 165.4 \\ 165.4 & 378.9 \end{bmatrix}$$

$$\underset{\sim}{W} = \begin{bmatrix} 400.6 & -197.5 \\ -197.5 & 472.6 \end{bmatrix}$$

89323.56

$$\underset{\sim}{W}^{-1} = \begin{bmatrix} 0.0031 & -0.0013 \\ -0.0013 & 0.0027 \end{bmatrix}$$

Now; $\underline{y} = \underline{a}' \underline{x}$

$\underline{a} = \underset{\sim}{\omega}^{-1}(\bar{x}_1 - \bar{x}_2)$ $6.213 - 7.412 = -1.199$

$$\underline{a} = \begin{bmatrix} 0.0031 & -0.0013 \\ -0.0013 & 0.0027 \end{bmatrix} \begin{bmatrix} -1.199 \\ -2.188 \end{bmatrix}$$

$\underline{a} = \begin{bmatrix} -0.0009 \\ -0.0043 \end{bmatrix}$, so the Fisher linear Discriminant function is.

$$y = \underline{a}' \underline{x}$$

$$\underline{y} = \begin{bmatrix} -0.0009 & -0.0043 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$\boxed{\underline{y} = -0.0009X_1 - 0.0043X_2}$$

Question No. 6:-

A class of 140 tenth grade children received four test, two open books $x_1' = [X_1 \ X_2]$ and two closed books $x_2' = [X_3 \ X_4]$. The mean vector and joint covariance matrix is given below so the discriminant rule is to allocate 'x' to the population 'P_1' if

$$y - \frac{1}{2}(\bar{y}_1 + \bar{y}_2) > 0$$ and to the population 'P_2' otherwise.

take $\bar{y} - \frac{1}{2}(\bar{y}_1 + \bar{y}_2)$

$$= \underline{a}'\underline{x} - \frac{1}{2}[\underline{a}'\bar{\underline{x}}_1 + a_2'\bar{\underline{x}}_2)$$

$$= a'\underline{x} - \frac{1}{2}[\underline{a}'(\bar{x}_1 + \bar{x}_2)]$$

$$\bar{y}_1 = a'\bar{\underline{x}}_1 = \begin{bmatrix} -0.0008946 & -0.004255 \end{bmatrix}\begin{bmatrix} 6.213 \\ 3.\ \ 33 \end{bmatrix} - 0.009$$

$$\bar{y}_1 = \;=\; -0.0188$$

$$\bar{y}_2 = a'\bar{\underline{x}}_2 = \begin{bmatrix} -0.0008946 & -0.004255 \end{bmatrix}\begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}$$

$$\bar{y}_2 = -0.02927$$

Allocation Rule:-

$$y - \frac{1}{2}(\bar{y}_1 + \bar{y}_2) > 0$$

$$[-0.0008946X_1 - 0.004255X_2] - \frac{1}{2}(-0.0188 - 0.02927) > 0$$

$$[-0.0008946X_1 - 0.004255X_2] - (-0.0240) > 0$$

$$-0.0008946X_1 - 0.004255X_2 + 0.0240 > 0$$

For $\underline{x} = [7.2 \quad 3.1]'$

consider; $-0.0008946(7.2) - 0.004255(3.1) + 0.0240 > 0$

$$0.004368 > 0$$

So the new observation $\underline{x} = [7.2 \quad 3.1]'$ will be allocated to the population "P_1".

$$\mu = \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 8 & 2 & 3 & 1 \\ 2 & 5 & -1 & 3 \\ 3 & -1 & 6 & -2 \\ 1 & 3 & -2 & 7 \end{bmatrix}$$

Find the canonical correlations between X_1 and X_2 and the first pair of canonical variables.

Solution:-

Given that

$$\underset{\sim}{\Sigma} = \left[\begin{array}{cc|cc} 8 & 2 & 3 & 1 \\ 2 & 5 & -1 & 3 \\ \hline 3 & -1 & 6 & -2 \\ 1 & 3 & -2 & 7 \end{array}\right] = \begin{bmatrix} \underset{\sim}{\Sigma}_{11} & \underset{\sim}{\Sigma}_{12} \\ \underset{\sim}{\Sigma}_{21} & \underset{\sim}{\Sigma}_{22} \end{bmatrix}$$

Now consider:

$$\underset{\sim}{\Sigma}_{11}^{-1} \underset{\sim}{\Sigma}_{12} \underset{\sim}{\Sigma}_{22}^{-1} \underset{\sim}{\Sigma}_{21} \longrightarrow (A)$$

$$\underset{\sim}{\Sigma}_{12} = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix} \quad \underset{\sim}{\Sigma}_{21} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$$

$$\underset{\sim}{\Sigma}_{11} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix} ; \quad \underset{\sim}{\Sigma}_{22} = \begin{bmatrix} 6 & -2 \\ -2 & 7 \end{bmatrix}$$

$$\underset{\sim}{\Sigma}_{11}^{-1} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 0.1389 & -0.0556 \\ -0.0556 & 0.2222 \end{bmatrix}$$

$$\underset{\sim}{\Sigma}_{22}^{-1} = \begin{bmatrix} 6 & -2 \\ -2 & 7 \end{bmatrix} = \begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1578 \end{bmatrix}$$

$$\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \begin{bmatrix} 0.1389^{\times 3+} & -0.0556(-1) \\ -0.0556 & 0.2222 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1578 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$$

$$\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \begin{bmatrix} 0.4723 & -0.027 \\ -0.3890 & 0.6110 \end{bmatrix}\begin{bmatrix} 0.6052 & -0.0264 \\ 0.3156 & 0.4208 \end{bmatrix}$$

$$\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \begin{bmatrix} 0.2770 & -0.0242 \\ -0.0426 & 0.2674 \end{bmatrix}$$

The characteristic equation is $|A - \lambda I| = 0$

$$\left|\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} - \lambda I\right| = 0$$

$$\left|\begin{bmatrix} 0.2770 & -0.0242 \\ -0.0426 & 0.2674 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right| = 0$$

$$\left|\begin{bmatrix} 0.2770 & -0.0242 \\ -0.0426 & 0.2674 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right| = 0$$

$$\begin{vmatrix} 0.2770 - \lambda & -0.0242 \\ -0.0426 & 0.2674 - \lambda \end{vmatrix} = 0$$

$$(0.2770 - \lambda)(0.2674 - \lambda) - 0.0010 = 0$$

$$0.0741 - 0.2770\lambda - 0.2674\lambda + \lambda^2 - 0.0010 = 0$$

$$\lambda^2 - 0.5444\lambda + 0.0731 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-0.5444) \pm \sqrt{(-0.5444)^2 - 4(1)(0.0731)}}{2(1)}$$

$$\lambda = \frac{0.5444 \pm \sqrt{0.2964 - 0.2924}}{2}$$

$$\lambda = \frac{0.5444 \pm 0.0632}{2}$$

$$\lambda_1 = \frac{0.5444 + 0.0632}{2} \quad ; \quad \lambda_2 = \frac{0.5444 - 0.0632}{2}$$

$$\lambda_1 = 0.3038 \quad ; \quad \lambda_2 = 0.2406$$

So the Canonical correlations are;

$$\Sigma_{v_1 v_2} = \sqrt{\lambda_1} = \sqrt{0.3038} = 0.5512$$

$$\Sigma_{v_1 v_2} = \sqrt{\lambda_2} = \sqrt{0.2406} = 0.4905$$

Now, the eigen vector corresponding to $\lambda_1 = 0.3038$

$$\left[\underset{\sim}{\Sigma}_{11}^{-1} \underset{\sim}{\Sigma}_{12} \underset{\sim}{\Sigma}_{22}^{-1} \underset{\sim}{\Sigma}_{21}\right] x = \lambda_1 x$$

$$\begin{bmatrix} 0.2770 & -0.0242 \\ 0.0426 & 0.2674 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.3038 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0.2770 x_1 - 0.0242 x_2 = 0.3038 x_1$$

$$0.2770 x_1 - 0.3038 x_1 - 0.0242 x_2 = 0$$

$$-0.0268 x_1 - 0.0242 x_2 = 0$$

$$0.0268 x_1 + 0.0242 x_2 = 0 \quad \text{———} \quad (1)$$

$$-0.0426 x_1 + 0.2674 x_2 = 0.3038 x_2$$

$$-0.0426 x_1 + 0.2674 x_2 - 0.3038 x_2 = 0$$

$$-0.0426 x_1 - 0.0364 x_2 = 0$$

$$0.0426 x_1 + 0.0364 x_2 = 0 \quad \text{———} \quad (2)$$

Adding eq ① and ②

$$0.0268 x_1 + 0.0242 x_2 = 0$$
$$0.0426 x_1 + 0.0364 x_2 = 0$$

$$0.0694 x_1 + 0.0606 x_2 = 0$$

Let $x_1 = 1$ then $x_2 = 0.8732$ $\quad \frac{1}{\sqrt{(1)^2 + (0.8732)^2}}$

$$\underline{x} = \begin{bmatrix} 1 \\ 0.8732 \end{bmatrix} \Rightarrow \underline{e}_1 = \begin{bmatrix} 0.7532 \\ 0.6577 \end{bmatrix};$$

Since, 'α_1' is subject to the constraint:

$$\alpha_1' \underset{\sim}{\Sigma}_{11} \alpha = 1$$

Take; $\alpha_1' \underset{\sim}{\Sigma}_{11} \alpha$

$$= \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 0.7532 \\ 0.6577 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} 7.3410 \\ 4.7949 \end{bmatrix}$$

$$= 5.5292 + 3.1536$$
$$= 8.6828 \neq 1.$$

Now, updated 'α_1' is.

$$\underline{\alpha_1} = \begin{bmatrix} 0.7532/\sqrt{8.6828} \\ 0.6577/\sqrt{8.6828} \end{bmatrix} = \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

Now,

$$\beta_1 = \lambda_1^{-1/2} \Sigma_{22}^{-1} \Sigma_{21} \underline{\alpha}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1578 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.6052 & -0.0264 \\ 0.3156 & 0.4208 \end{bmatrix} \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.1488 \\ 0.1746 \end{bmatrix}$$

$$\beta_1 = 1.8143 \begin{bmatrix} 0.1488 \\ 0.1746 \end{bmatrix} = \begin{bmatrix} 0.2700 \\ 0.3168 \end{bmatrix}$$

Now; the first pair of canonical variates is;

$$u_1 = \underline{\alpha}_1' X_1 = \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$u_1 = 0.7532x_1 + 0.6577x_2$$

$$v_1 = \underline{\beta}_1' X_2 = \begin{bmatrix} 0.2700 & 0.3168 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$$

$$v_1 = 0.2700x_3 + 0.3168x_4$$

Hence, the two variates which are maximizing the correlation between two sets of variables (x_1, x_2) and (x_3, x_4) are 'u_1' and 'v_1' and the maximum correlation is : 0.3168.

Question No. 5:-

The following mean vectors and covariance matrices are based on $n_1 = n_2 = 100$ observations.

$$\bar{x}_1 = \begin{bmatrix} 6.213 \\ 3.133 \end{bmatrix}, \quad \bar{x}_2 = \begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}; \quad S_1 = \begin{bmatrix} 1.813 & 0.32 \\ 0.32 & 0.935 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 2.193 & 1.654 \\ 1.654 & 3.789 \end{bmatrix}$$

Find the Fisher's Linear Discriminant function and Discriminant rule. Also allocate the new observations $\bar{x}' = [7.2 \quad 3.3]$ to any of these populations.

Solution:-

Repeat.

Question No. 6:-

A class of 140 tenth grade children received four test, two open books $x_1' = [X_1 \quad X_2]$ and two closed books $x_2' = [X_3 \quad X_4]$. The mean vector and joint covariance matrix is given below:

-: Past Paper 2015:-

Question No. 5:-

The following mean vector and covariance matrices are based on $n_1 = n_2 = 100$ observations.

$$\bar{X}_1 = \begin{bmatrix} 6.213 \\ 3.133 \end{bmatrix}, \bar{X}_2 = \begin{bmatrix} 7.412 \\ 5.321 \end{bmatrix}, S_1 = \begin{bmatrix} 1.813 & 0.32 \\ 0.32 & 0.985 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 2.193 & 1.654 \\ 1.654 & 3.789 \end{bmatrix}$$

Find the Fisher's Linear Discriminant function and Discriminant rule. Also allocate the new observations $\bar{x}' = [7.2 \quad 3.1]$ to any of these populations.

Solution:-

Repeat.

Question No. 6:-

A class of 140 tenth grade children received four test, two open books $X_1' = [X_1 \quad X_2]$ and two closed books $X_2' = [X_3 \quad X_4]$. The mean vector and joint covariance matrix is given below:

$$\mu = \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 8 & 2 & 3 & 1 \\ 2 & 5 & -1 & 3 \\ 3 & -1 & 6 & -2 \\ 1 & 3 & -2 & 7 \end{bmatrix}$$

Find the cannonical correlations between X_1 and X_2 and the first pair of cannonical variables. Compute the correlations between the first pair of cannonical variates and their component variables.

Solution:-

Repeat

Question No. 3:-

Data for two variables give the summary:

$$n = 42, \quad \bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}, \quad S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

Test the Hypothesis $H_0: \mu = [0.562 \quad 0.589]'$ find 95% confidence ellipse for μ consisting of all values (μ_1, μ_2). Also draw the ellipse.

Solution:-

(a) Confidence Region / Ellipse :-

As;

$$\underset{\sim}{S} = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

$$\underset{\sim}{S}^{-1} = \begin{bmatrix} 199.0457 & -159.5092 \\ -159.5092 & 196.3190 \end{bmatrix}$$

also;

$$c^2 = \left(\frac{vp}{v-p+1}\right) F_{0.05}(p, v-p+1) \qquad \because v = n-1 \quad \because n = 42$$

$$c^2 = \frac{(42-1)(2)}{(42-1-2+1)} F_{0.05}(2, 42-2 = 40) \qquad \because p = 2$$

After

$$c^2 = (2.05)(3.23)$$

$$c^2 = 6.6215$$

The 95% Confidence Ellipse is :-

$$n(\bar{x} - \underline{\mu})' \underset{\sim}{S}^{-1} (\bar{x} - \underline{\mu}) \leq c^2$$

$$42 \left(\begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \right)' \begin{bmatrix} 199.0457 & -159.5092 \\ -159.5092 & 196.3190 \end{bmatrix} \left(\begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \right) \leq 6.6215$$

$$42 \begin{bmatrix} 0.564 - \mu_1 & 0.603 - \mu_2 \end{bmatrix} \begin{bmatrix} 199.0457 & -159.5092 \\ -159.5092 & 196.3190 \end{bmatrix} \begin{bmatrix} 0.564 - \mu_1 \\ 0.603 - \mu_2 \end{bmatrix} \leq 6.6215$$

$\rightarrow$ A

The center of the confidence ellipse will be at

$$\bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}$$

Now for directions and lengths:

$$S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}.$$

The characteristic equation is

$$|S - \lambda I| = 0$$

$$\left| \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 0.0144 - \lambda & 0.0117 \\ 0.0117 & 0.0146 - \lambda \end{vmatrix} = 0$$

$$(0.0144 - \lambda)(0.0146 - \lambda) - (0.0117)^2 = 0$$

$$0.00021024 - 0.0144\lambda - 0.0146\lambda + \lambda^2 - 0.00013689 = 0$$

$$\lambda^2 - 0.029\lambda + 0.00007335$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-0.029) \pm \sqrt{(-0.029)^2 - 4(1)(0.00007335)}}{2(1)}$$

$$\lambda = \frac{0.029 \pm 0.023400855}{2}$$

$$\lambda_1 = \frac{0.029 + 0.0234}{2}, \quad \lambda_2 = \frac{0.029 - 0.0234}{2}$$

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$\lambda_1 = 0.0262$; $\lambda_2 = 0.0028$

Now Eigen vectors:

$$S\, x = \lambda_1 x$$

$$\begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.0262 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0.0144 x_1 + 0.0117 x_2 = 0.0262 x_1$$

$$0.0144 x_1 + 0.0117 x_2 - 0.0262 x_1 = 0$$

$$-0.0118 x_1 + 0.0117 x_2 = 0$$

$$0.0118 x_1 - 0.0117 x_2 = 0$$

$$0.0117 x_1 + 0.0146 x_2 = 0.0262 x_2 = 0$$

$$0.0117 x_1 + 0.0146 x_2 - 0.0262 x_2 = 0$$

$$0.0117 x_1 - 0.0116 x_2 = 0$$

let

$$x_1 = 1 \;,\; x_2 = 1$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow e_1 = \begin{bmatrix} 1/\sqrt{1^2+1^2} \\ 1/\sqrt{1^2+1^2} \end{bmatrix} = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$$

Now for λ_2:

$$S\, x = \lambda_2 x$$

$$\begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.0028\,x_1 \\ 0.0028\,x_2 \end{bmatrix}$$

$\rightarrow$ $0.0144x_1 + 0.0117x_2 = 0.0028x_1$

$0.0144x_1 + 0.0117x_2 - 0.0028x_1 = 0$

$0.0116x_1 + 0.0117x_2 = 0$

$\rightarrow$ $0.0117x_1 + 0.0146x_2 = 0.0028x_2$

$0.0117x_1 + 0.0146x_2 - 0.0028x_2 = 0$

$0.0117x_1 + 0.0118x_2 = 0$

let $x_1 = 1$; $x_2 = -1$

$$\underline{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow e_2 = \begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$$

Length of Major Axis:-

$$= 2\sqrt{\frac{\lambda_1 c^2}{n}} = 2\sqrt{\frac{(0.0262)(6.6215)}{42}} = 0.1285$$

Length of Minor Axis:-

$$= 2\sqrt{\frac{\lambda_2 c^2}{n}} = 2\sqrt{\frac{(0.0028)(6.6215)}{42}} = 0.0420$$

Directions:- The major axis will lie in the direction of $\underline{e_1} = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$ and the minor axis will lie in the in the direction of $\underline{e}_2 = \begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix}$

To test:

$H_0: \mu = \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$

$H_1: \mu \neq \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$

substitute this value in eq Ⓐ, we get

$$(42)\begin{bmatrix} 0.002 & 0.014 \end{bmatrix}\begin{bmatrix} 199.04 & -159.509 \\ -159.509 & 196.319 \end{bmatrix}\begin{bmatrix} 0.002 \\ 0.014 \end{bmatrix} \leq 6.6215$$

$$(42)\begin{bmatrix} 0.002 & 0.014 \end{bmatrix}\begin{bmatrix} -1.8350 \\ 2.4294 \end{bmatrix} \leq 6.6215$$

$$(42)(0.0302) \leq 6.6215$$

$$1.274 \leq 6.6215$$

Hence $H_0: \underline{\mu} = \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$ will not be rejected.

:- Past Paper 2017 :-

All Long Questions repeated.

:- Past Paper 2016:

All Long Questions repeated

:- Past Paper 2019:-

Question No. 3 and Question No. 6 repeat.

Question No. 5:-

Suppose 'x' is multinomial random variable which comes either from P_1 with multinomial probabilities $\alpha_1, \alpha_2, \ldots, \alpha_k$ or from P_2 with multinomial probabilities $\beta_1, \beta_2, \ldots, \beta_k$ where;

$$\sum_{i=1}^{k} \alpha_i = 1, \quad \sum_{i=1}^{k} \beta_i = 1 \quad \text{and} \quad \sum_{i=1}^{k} x_i = n$$

Obtain the Maximum Likelihood Discriminant Rule.

Solution:-

Since 'P_1' is Multinomial with probabilities $\alpha_1, \alpha_2, \ldots, \alpha_k$; So;

$$f_1(\underline{x}) = \frac{n!}{x_1!, x_2! \ldots x_k!} (\alpha_1)^{x_1} (\alpha_2)^{x_2} \ldots (\alpha_k)^{x_k}$$

$$f_1(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} \prod_{i=1}^{k} (\alpha_i)^{x_i}$$

Similarly;

$$f_2(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} (\beta_1)^{x_1} \cdot (\beta_2)^{x_2} \cdots (\beta_k)^{x_k}$$

$$f_2(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} \prod_{i=1} (\beta_i)^{x_i}$$

Then;
$$\lambda = \frac{f_1(\underline{x})}{f_2(\underline{x})} = \frac{\prod_{i=1}^{k} (\alpha_i)^{x_i}}{\prod_{i=1}^{k} (\beta_i)^{x_i}}$$

$$\lambda = \prod_{i=1}^{k} \left(\frac{\alpha_i}{\beta_i}\right)^{x_i}$$

Taking log on both sides, we get;

$$\ln \lambda = \sum_{i=1}^{k} \ln \left(\frac{\alpha_i}{\beta_i}\right)^{x_i}$$

$$\ln \lambda = \sum_{i=1}^{k} x_i \ln \left(\frac{\alpha_i}{\beta_i}\right)$$

The Maximum likelihood discriminant Rule is to allocate '$\underline{x}$' to P_1 if $\lambda \geq 1$ and to P_2 otherwise. This could also be written as;

'x' will be allocated to P_1 if $\ln \lambda > 0$ and to P_2 otherwise.

Question No. 4:-

$$f_1(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} \prod_{i=1}^{k} (\alpha_i)^{x_i}$$

Similarly;

$$f_2(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} (\beta_1)^{x_1} \cdot (\beta_2)^{x_2} \cdots (\beta_k)^{x_k}$$

$$f_2(\underline{x}) = \frac{n!}{x_1!\, x_2! \cdots x_k!} \prod_{i=1}^{k} (\beta_i)^{x_i}$$

Then; $\lambda = \dfrac{f_1(x)}{f_2(x)} = \dfrac{\prod_{i=1}^{k} (\alpha_i)^{x_i}}{\prod_{i=1}^{k} (\beta_i)^{x_i}}$

$$\lambda = \prod_{i=1}^{k} \left(\frac{\alpha_i}{\beta_i}\right)^{x_i}$$

Taking log on both sides, we get;

$$\ln \lambda = \sum_{i=1}^{k} \ln \left(\frac{\alpha_i}{\beta_i}\right)^{x_i}$$

$$\ln \lambda = \sum_{i=1}^{k} x_i \ln \left(\frac{\alpha_i}{\beta_i}\right)$$

The Maximum likelihood discriminant Rule is to allocate '$\underline{x}$' to P_1 if $\lambda \geq 1$ and to P_2 otherwise. This could also be written as;

'x' will be allocated to P_1 if $\ln \lambda > 0$ and to P_2 otherwise.

Question No. 4:-

Discuss the principal components from equal correlation matrix

$$\rho = \begin{bmatrix} 1 & 0.6 & 0.6 \\ 0.6 & 1 & 0.6 \\ 0.6 & 0.6 & 1 \end{bmatrix}$$

Solution:-

$$\rho = \begin{bmatrix} 1 & 0.6 & 0.6 \\ 0.6 & 1 & 0.6 \\ 0.6 & 0.6 & 1 \end{bmatrix}$$

$$\bar{r} = \frac{\bar{r}_1 + \bar{r}_2 + \bar{r}_3}{3}$$

$$\bar{r}_k = \frac{1}{p-1} \sum_{i=1}^{3} r_{i1}$$

$$\bar{r}_1 = \frac{1}{3-1}(0.6 + 0.6) = 0.6$$

$$\bar{r}_2 = \frac{1}{3-1}(0.6 + 0.6) = 0.6$$

$$\bar{r}_3 = \frac{1}{3-1}(0.6 + 0.6) = 0.6$$

$$\bar{r} = \frac{0.6 + 0.6 + 0.6}{3} = 0.6$$

$$\lambda_1 = 1 + (p-1)\bar{r}$$

$$\lambda_1 = 1 + (3-1)(0.6)$$

$$\lambda_1 = 2.20$$

$$\lambda_2 = \lambda_3 = 1 - \bar{r} = 1 - 0.6 = 0.4$$

$$\underline{e}_1 = \left[\frac{1}{\sqrt{p}} \quad \frac{1}{\sqrt{p}} \quad \frac{1}{\sqrt{p}}\right]'$$

$$\underline{e}_1 = \left[\frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}}\right]'$$

$$\underline{e}_1 = [0.5774 \quad 0.5774 \quad 0.5774]'$$

$$\underline{e}_2 = \left[\frac{1}{\sqrt{1\times2}} \quad \frac{-1}{\sqrt{1\times2}} \quad 0\right]'$$

$$\underline{e}_2 = [0.7071 \quad -0.7071 \quad 0]'$$

$$\underline{e}_3 = \left[\frac{1}{\sqrt{2\times3}} \quad \frac{1}{\sqrt{2\times3}} \quad \frac{-2}{\sqrt{2\times3}}\right]'$$

$$\underline{e}_3 = [0.4082 \quad 0.4082 \quad -0.8165]'$$

(ii) PC's:

PC's are:

$$Y_1 = \underline{e}_1' \underline{Z} = [0.5774 \quad 0.5774 \quad 0.5774] \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \end{bmatrix}$$

$$Y_1 = 0.5774 Z_1 + 0.5774 Z_2 + 0.5774 Z_3$$

$$Y_2 = \underline{e}_2' \underline{Z} = [0.7071 \quad -0.7071 \quad 0] \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \end{bmatrix}$$

$$Y_2 = 0.7071 z_1 - 0.7071 z_2$$

$$Y_3 = e_3' z = [0.4082 \quad 0.4082 \quad -0.8165] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

$$Y_3 = 0.4082 z_1 + 0.4082 z_2 - 0.8165 z_3$$

(ii) PEV:-

$$\text{PEV by PC-I} = \frac{\lambda_1}{\lambda_1+\lambda_2+\lambda_3} \times 100 = \frac{2.20}{2.20+0.6+0.6} \times 100$$

$$= 64.70\%$$

$$\text{PEV by PC-II} = \frac{\lambda_2}{\lambda_1+\lambda_2+\lambda_3} = \frac{0.6}{2.20+0.6+0.6} \times 100$$

$$= 17.65\%$$

$$\text{PEV by PC-III} = \frac{\lambda_3}{\lambda_1+\lambda_2+\lambda_3} = \frac{0.6}{2.20+0.6+0.6} \times 100$$

$$= 17.65\%$$

-: Past Paper 2015 :-

Question No. 4:- 2021

Let X_1 and X_2 be two random variables with covariance matrix.

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

i- Obtain the Principal Components and find the percentage of variation explained by each.

ii- Change Σ into correlation matrix ρ. And find Principal components using ρ.

iii- Are Principal Components from (i) and (ii) same? If yes, why?

Solution:-

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

i- Principal Components:-

The characteristic equation is:

$$|\Sigma - \lambda I| = 0$$

$$\left|\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right| = 0$$

$$\left|\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right| = 0$$

$$\left|\begin{bmatrix} 9-\lambda & \sqrt{6} \\ \sqrt{6} & 4-\lambda \end{bmatrix}\right| = 0$$

$$(9-\lambda)(4-\lambda) - (\sqrt{6})^2 = 0$$

$$36 - 9\lambda - 4\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 - 13\lambda + 30 = 0$$

$$\lambda = \frac{-(-13) \pm \sqrt{(-13)^2 - 4(1)(30)}}{2(1)}$$

$$\lambda = \frac{13 \pm \sqrt{169 - 120}}{2} \quad ; \quad \lambda = \frac{13 \pm \sqrt{49}}{2}$$

$$\lambda = \frac{13 \pm 7}{2}$$

$$\lambda_1 = \frac{13+7}{2} \quad ; \quad \lambda_2 = \frac{13-7}{2}$$

$$\boxed{\lambda_1 = 10} \quad ; \quad \boxed{\lambda_2 = 3}$$

Eigen vector for $\lambda_1 = 10$

$$\Sigma x = \lambda_1 x$$

$$\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 10\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$9x_1 + \sqrt{6}\,x_2 = 10x_1$$

$$9x_1 + \sqrt{6}\,x_2 - 10x_1 = 0$$

$$-x_1 + \sqrt{6}\,x_2 = 0 \quad \text{———} \quad (i)$$

$$\sqrt{6}\,x_1 + 4x_2 = 10x_2$$
$$\sqrt{6}\,x_1 + 4x_2 - 10x_2 = 0$$
$$\sqrt{6}\,x_1 - 6x_2 = 0 \longrightarrow (2)$$

let $x_1 = 1$ put in (1).

$$-1 + \sqrt{6}\,x_2 = 0$$
$$x_2 = 1/\sqrt{6} \Rightarrow \boxed{x_2 = 0.4082}$$

$$x_1 = \begin{bmatrix} 1 \\ 0.4082 \end{bmatrix} \Rightarrow \underline{e}_1 = \begin{bmatrix} 1/\sqrt{1^2+0.4082^2} \\ 0.4082/\sqrt{1^2+0.4082^2} \end{bmatrix}$$

$$\underline{e}_1 = \begin{bmatrix} 0.9258 \\ 0.3779 \end{bmatrix}$$

Eigen vector for $\lambda_2 = 3$

$$\Sigma x = \lambda_2 x$$

$$\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$9x_1 + \sqrt{6}\,x_2 = 3x_1$$
$$9x_1 - 3x_1 + \sqrt{6}\,x_2 = 0$$
$$6x_1 + \sqrt{6}\,x_2 = 0 \quad \text{---} \quad (3)$$
$$\sqrt{6}\,x_1 + 4x_2 = 3x_2$$
$$\sqrt{6}\,x_1 + 4x_2 - 3x_2 = 0$$
$$\sqrt{6}\,x_1 + x_2 = 0 \quad \text{---} \quad (4)$$

let $x_1 = 1$ in (3)

$$6 + \sqrt{6}\,x_2 = 0$$
$$x_2 = \frac{-6}{\sqrt{6}} = -2.4495$$

$$X = \begin{bmatrix} 1 \\ -2.4495 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 1/\sqrt{1^2+(-2.4495)^2} \\ -2.4495/\sqrt{1^2+(-2.4495)^2} \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.3780 \\ -0.9258 \end{bmatrix}$$

Principal Components:-

$$1^{st}\ PC = Y_1 = \underline{e_1'}\,\underline{x} = \begin{bmatrix} 0.9258 & 0.3779 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_1 = 0.9258x_1 + 0.3779x_2$$

$$2^{nd}\ PC = Y_2 = \underline{e_2'}\,\underline{x} = \begin{bmatrix} 0.3780 & -0.9258 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_2 = 0.3780x_1 - 0.9258x_2$$

Percentage of variation explained:-

%age of explained variation by PC-I:

$$= \frac{\lambda_1}{\lambda_1+\lambda_2} \times 100$$

$$= \frac{10}{10+3} \times 100 = 76.92\%$$

%age of explained Variation by PC-II:-

$$= \frac{\lambda_2}{\lambda_1+\lambda_3} \times 100 = \frac{3}{10+3} \times 100$$

$$= 23.077\%$$

ii- Change Σ into correlation matrix ρ

Find . P.C using ρ.

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

We know that

$$\rho = \underset{\sim}{D}^{-1/2} \underset{\sim}{\Sigma} \underset{\sim}{D}^{-1/2}$$

$$D = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix} = D^{-1/2} \begin{bmatrix} (9)^{-1/2} & 0 \\ 0 & (4)^{-1/2} \end{bmatrix}$$

$$D^{-1/2} = \begin{bmatrix} 0.3333 & 0 \\ 0 & 0.5000 \end{bmatrix}$$

$$\rho = \begin{bmatrix} 0.3333 & 0 \\ 0 & 0.5000 \end{bmatrix} \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} \begin{bmatrix} 0.3333 & 0 \\ 0 & 0.5000 \end{bmatrix}$$

$$\underset{\sim}{\rho} = \begin{bmatrix} 0.3333 & 0 \\ 0 & 0.5000 \end{bmatrix} \begin{bmatrix} 2.9997 & 1.2247 \\ 0.8164 & 2.0000 \end{bmatrix}$$

$$\underset{\sim}{\rho} = \begin{bmatrix} 0.9998 & 0.4082 \\ 0.4082 & 1.0000 \end{bmatrix}$$

Principal Components:-

$$|\underset{\sim}{\rho} - \lambda \underset{\sim}{I}| = 0$$

$$\left| \begin{bmatrix} 0.9998 & 0.4082 \\ 0.4082 & 1.000 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 0.9998 & 0.4082 \\ 0.4082 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 0.9998-\lambda & 0.4082 \\ 0.4082 & 1-\lambda \end{vmatrix} = 0$$

$$(0.9998-\lambda)(1-\lambda) - (0.4082)^2 = 0$$

$$0.9998 - 0.9998\lambda - \lambda + \lambda^2 - 0.1666 = 0$$

$$\lambda^2 - 1.9998\lambda + 0.8332 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-1.9998) \pm \sqrt{(-1.9998)^2 - 4(1)(0.8332)}}{2(1)}$$

$$\lambda = \frac{1.9998 \pm 0.8163}{2}$$

$$\lambda_1 = \frac{1.9998 + 0.8163}{2} \quad ; \quad \lambda_2 = \frac{1.9998 - 0.8163}{2}$$

$$\lambda_1 = 1.4081 \quad ; \quad \lambda_2 = 0.5918$$

Eigen vector for $\lambda_1 = 1.4081$

$$\underset{\sim}{\rho}\, \underline{x} = \lambda_1 \underline{x}$$

$$\begin{bmatrix} 0.9998 & 0.4082 \\ 0.4082 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1.4081 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0.9998x_1 + 0.4082x_2 = 1.4081x_1$$

$$0.9998x_1 - 1.4081x_1 + 0.4082x_2 = 0$$

$$-0.4083x_1 - 0.4082x_2 = 0$$

$$0.4083x_1 + 0.4082x_2 = 0 \quad \text{———} (1)$$

$$0.4082x_1 + x_2 = 1.4082x_2$$

$$0.4082x_1 + x_2 - 1.4082x_2 = 0$$

$$0.4082x_1 - 0.4082x_2 = 0 \longrightarrow (2)$$

let $x_1 = 1$ in(1) then

$$0.4083(1) - 0.4082x_2 = 0$$

$$x_2 = -\frac{0.4083}{-0.4082}$$

$$x_2 = 1.0002$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.0002 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1/\sqrt{1^2 + 1.0002^2} \\ 1.0002/\sqrt{1^2 + 1.0002^2} \end{bmatrix} = \begin{bmatrix} 0.7070 \\ 0.7072 \end{bmatrix}$$

Eigen vector for $\lambda_2 = 0.5918$

$$\underset{\sim}{e}\, \underline{x} = \lambda_2 \underline{x}$$

$$\begin{bmatrix} 0.9998 & 0.4082 \\ 0.4082 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.5918 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0.9998x_1 + 0.4082x_2 = 0.5918x_1$$

$$0.9998x_1 - 0.5918x_1 + 0.4082x_2 = 0$$

$$0.4080x_1 + 0.4082x_2 = 0 \text{ ——— (i)}$$

$$0.4082x_1 + x_2 = 0.5918x_2$$

$$0.4082x_1 + x_2 - 0.5918x_2 = 0$$

$$0.4082x_1 + 0.4082x_2 = 0 \text{ ——— (ii)}$$

let $x_1 = 1$ in (1)

$$0.4080 + 0.4082 x_2 = 0$$

$$x_2 = \frac{-0.4080}{0.4082}$$

$$x_2 = -0.9995$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.9995 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 1/\sqrt{1^2 + (-0.9995)^2} \\ -0.9995/\sqrt{1^2 + (-0.9995)^2} \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.7073 \\ -0.7069 \end{bmatrix}$$

Principal Components:-

$$1^{st}\ PC = Y_1 = e_1' X = \begin{bmatrix} 0.7070 & 0.7072 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_1 = 0.7070 x_1 + 0.7072 x_2$$

$$2^{nd}\ PC = Y_2 = e_2' X = \begin{bmatrix} 0.7073 & -0.7069 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_2 = 0.7073 x_1 - 0.7069 x_2$$

Percentage of Explained variation:-

% age of E.V by PC - I:-

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100 = \frac{1.4081}{1.4081 + 0.5918} \times 100 = 70.41\%$$

% age of E.V by PC II :-

$$= \frac{\lambda_2}{\lambda_1 + \lambda_2} \times 100$$

$$= \frac{0.5918}{1.4081 + 0.5918} \times 100$$

$$= 29.5915\%$$

No, The P.C's from ① and ② are not same.

$tr(\Sigma) = \sigma_{11} + \sigma_{22} \dots \sigma_{pp}$

$\Sigma = P \Lambda P'$ where $\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_p \end{bmatrix}$

$\underset{\sim}{P} = [e_1 \; e_2 \dots e_p]$

$tr(\Sigma) = tr(P \Lambda P')$ P is orthogonal

$tr(\Sigma) = tr(\Lambda P' P)$

$tr(\Sigma) = tr(\Lambda I)$ $\quad P'P = PP' = I$

$tr(\Sigma) = tr(\Lambda)$

$tr(\Sigma) = \lambda_1 + \lambda_2 + \dots \lambda_p = Var(Y_1) + Var Y_2 + \dots Var(Y_p)$

$tr(\Sigma) = \sum_{i=1} Var\ Y_i$

From eq ① and ②

$\sigma_{11} + \sigma_{22} \dots + \sigma_{pp} = Var\ X_1 + \dots Var(X_p) = \sum_{i=1}^{p} Var(X_i)$

$\sum_{i=1}^{p} Var(Y_i)$